

*"99.9% of science is boring."
- Makise Kurisu*

PRECALCULUS

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CONTENTS

1	Functions	1
1.1	Intro To Functions	1
1.2	Evaluating Functions	3
1.2.1	Difference Quotient	4
1.3	Finding Domains	5
1.3.1	The Problem Areas	5
1.4	Operations Of Functions	7
1.5	The Vertical Line Test	8
1.6	Features of Graphs Points, Domain, Range	9
1.7	Properties of Functions: Even or Odd	11
1.8	Properties of Functions: Increasing or Decreasing	13
1.9	Properties of Functions: Extremas	15
1.10	Average Rate of Change	16
1.11	Piece-wise Functions	17
1.12	Important Functions	19
2	Graphical Transformations	22
2.1	Introduction To Graphical Transformations	22
2.1.1	Vertical	22
2.1.2	Horizontal	22
2.1.3	Vertical Stretch/Compression	24
2.1.4	Horizontal Stretch/Compression	24
2.1.5	Reflection	25
2.2	How to Graph with Transformations	27
3	Quadratics	29
3.1	Intro to Solving Quadratics	29
3.1.1	Cases	29
3.2	The Square Root Method for Quadratics	31
3.3	Factoring Method for Quadratics	33
3.4	Completing the Square Method for Quadratics	36
3.5	Proving the Quadratic Formula	39
3.5.1	Proof	39
3.6	The Quadratic Formula	40
4	Intersection of Functions	42

5 Solving Equations by Using Substitutions	44
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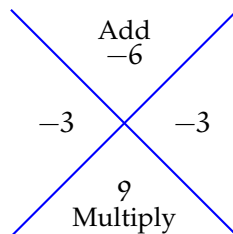
5 SOLVING EQUATIONS BY USING SUBSTITUTIONS

Note: also called u-substitution.

Definition 5.1 (u-substitution). Solves equations of quadratic form by rewriting them as $a(g(x))^2 + b(g(x)) + c = 0$ where $u = g(x)$ transforms the equation into the standard quadratic form $au^2 + bu + c = 0$.

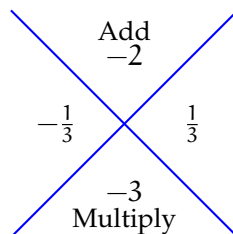
Example 5.1 (Equations).

$$\begin{aligned} 1. \quad & g(x) = (3x + 4)^2 - 6(3x + 4) + 9 \\ & (3x + 4)^2 - 6(3x + 4) + 9 = 0 \\ & u = 3x + 4 \\ & u^2 - 6u + 9 = 0 \end{aligned}$$



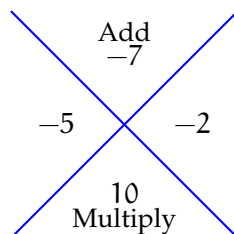
$$\begin{aligned} & (u - 3)(u - 3) = 0 \\ & (u - 3)^2 = 0 \\ & (u - 3) = 0 \\ & u = 3 \rightarrow 3x + 4 = 3 \rightarrow 3x + 4 = 3 \\ & 3x = -1 \\ & x = -\frac{1}{3} \end{aligned}$$

$$\begin{aligned} 2. \quad & f(x) = 3x^4 - 2x^2 - 1 \\ & 3x^4 - 2x^2 - 1 = 0 \\ & 3(x^2)^2 - 2(x^2) - 1 = 0 \\ & u = x^2 \\ & 3u^2 - 2u - 1 = 0 \end{aligned}$$



$$\begin{aligned} & u = 1 \wedge u = -\frac{1}{3} \\ & (u - 1)(3u + 1) = 0 \\ & u - 1 = 0 \wedge 3u + 1 = 0 \\ & x^2 = 1 \wedge x^2 = -\frac{1}{3} \\ & \sqrt{x^2} = \pm\sqrt{1} \wedge \sqrt{x^2} = \pm\sqrt{-\frac{1}{3}} \\ & (x = 1 \wedge x = i\sqrt{\frac{1}{3}}) \wedge (x = -1 \wedge x = -i\sqrt{\frac{1}{3}}) \end{aligned}$$

$$\begin{aligned}
 3. \quad & f(x) = x^{\frac{2}{3}} - 7x^{\frac{1}{3}} + 10 \\
 & x^{\frac{2}{3}} - 7x^{\frac{1}{3}} + 10 = 0 \\
 & (x^{\frac{1}{3}})^2 - 7(x^{\frac{1}{3}}) + 10 = 0 \\
 & u = x^{\frac{1}{3}} \\
 & u^2 - 7u + 10 = 0
 \end{aligned}$$

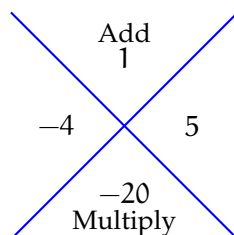


$$\begin{aligned}
 (u - 5)(u - 2) &= 0 \\
 u - 5 = 0 \wedge u - 2 &= 0 \\
 u = 5 \wedge u &= 2 \\
 x^{\frac{1}{3}} = 5 \wedge x^{\frac{1}{3}} &= 2
 \end{aligned}$$

Note: We can do two methods here.

$$\begin{aligned}
 \sqrt[3]{x} &= 5 \\
 (\sqrt[3]{x})^3 &= (5)^3 \\
 x &= 125 \\
 x^{\frac{1}{3}} &= 2 \\
 (x^{\frac{1}{3}})^3 &= (2)^3 \\
 x &= 8
 \end{aligned}$$

$$\begin{aligned}
 4. \quad & g(x) = x + \sqrt{x} - 20 \\
 & x + \sqrt{x^2} - 20 = 0 \\
 & x^1 + x^{\frac{1}{2}} - 20 = 0 \\
 & (x^{\frac{1}{2}})^2 + (x^{\frac{1}{2}}) - 20 = 0 \\
 & u = x^{\frac{1}{2}} \\
 & u^2 + u - 20 = 0
 \end{aligned}$$



$$\begin{aligned}
 u = 4 \wedge u &= -5 \\
 x^{\frac{1}{2}} = 4 \wedge x^{\frac{1}{2}} &= -5 \\
 \sqrt{x} = 4 \wedge \sqrt{x} &= -5 \\
 (\sqrt{x})^2 &= (4)^2 \wedge (\sqrt{x})^2 = (-5)^2 \\
 x = 16 \wedge x &= 25
 \end{aligned}$$

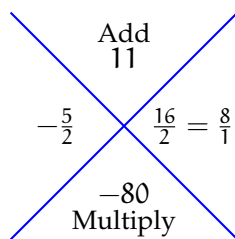
Note: $x = 25$ is not a real solution because a square root can't equal a negative.

$$\begin{aligned}
 5. \quad & h(x) = -2y^{-2} - 11y^{-1} + 40 \\
 & -2y^{-2} - 11y^{-1} + 40 = 0 \\
 & 2y^{-2} + 11y^{-1} - 40 = 0
 \end{aligned}$$

$$2(y^{-1})^2 + 11(y^{-1}) - 40 = 0$$

$$u = y^{-1}$$

$$2u^2 + 11u - 40 = 0$$



Add
11

$-\frac{5}{2}$ $\frac{16}{2} = \frac{8}{1}$

-80
Multiply

$$(2u - 5)(u + 8) = 0$$

$$2u - 5 = 0 \wedge u + 8 = 0$$

$$u = \frac{5}{2} \wedge u = -8$$

$$y^{-1} = \frac{5}{2} \wedge y^{-1} = -8$$

$$\frac{1}{y} = \frac{5}{2} \wedge \frac{1}{y} = -8$$

$$y = \frac{2}{5} \wedge y = -\frac{1}{8}$$